A defining polynomial in twenty-seven variables: an existence theorem with a uniform degree bound

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Abstract

Under the four arithmetic inputs of the specified elliptic source, we prove that the integers are the rational input projection of an integer polynomial in twenty-seven total variables and of degree less than 2^{87}. We give a computable family F_L of such polynomial specifications, indexed by a fixed odd integer coefficient parameter L. Every sufficiently large L has the required input projection. The proof uses von Buhren's uniform Mordell–Lang height theorem; its ineffective additive constant prevents us from certifying a numerical sufficient value of L. Thus the variable and degree bounds are proved, while no particular member is asserted here to be a fully explicit defining polynomial. Exact source, finite-algebra, and degree certificates accompany the family.

1 Statement and arithmetic inputs

All variable counts include the distinguished input. A polynomial $F \in \mathbb{Z}[X, Y_1, \dots, Y_m]$ defines $\mathbb{Z}$ over $\mathbb{Q}$ if

$$x \in \mathbb{Z} \quad \Longleftrightarrow \quad \exists \mathbf{y} \in \mathbb{Q}^m \quad F(x, \mathbf{y}) = 0 \qquad (x \in \mathbb{Q}).$$

The integer L below is fixed when a polynomial is constructed; it is neither its input nor an existential variable.

Theorem 1. *Assume the elliptic group and height assertion, the exponent nineteen contact-height bound, the pole obstruction and simultaneous-realization input, and the simultaneous prime-pattern input specified in Section 1.1. There is a computable family*

$$F_L \in \mathbb{Z}[X, Y_1, \dots, Y_{26}], \qquad L \geq 3 \text{ odd},$$

of finite polynomial specifications with the following properties.

1. *Every member has degree less than 2^{87}.*
2. *There is a fixed real number L_0 such that every odd $L \geq L_0$ defines $\mathbb{Z}$ over $\mathbb{Q}$.*
3. *The proof does not determine an effective sufficient value of L_0, and does not give an effective coefficient-height bound for a member known to define $\mathbb{Z}$.*

The finite specification uses integer coefficient arithmetic, determinants, resultants, and finite products, all with explicit finite index ranges. It uniquely specifies an ordinary polynomial, without treating the matrix coordinates or algebra generators as quantified variables.

1.1 The four inherited inputs

Set

$$K = \mathbb{Q}(\sqrt{2}), \quad d = 3 - \sqrt{2}, \quad D = d^2 = 11 - 6\sqrt{2}, \quad E : y^2 = x^3 - Dx,$$
$$P = \left(\frac{57}{8} - \frac{7}{2}\sqrt{2}, -\frac{165}{8} + \frac{417}{32}\sqrt{2}\right).$$

The construction uses the following assertions, with exactly the scope established in the arithmetic dependency paper [4].

1. $2E(K) = \mathbb{Z}P$, and $h(x([n]P)) \geq n^2 - 6$ for every nonzero integer n.

2. For the five normalized contacts with $\infty, 0, 1, 2401, -343/32$, the odd contact radical $M_5(q)$ outside primes at most 76176 satisfies $h(q) \leq HM_5(q)^{19}$ when q is not a contact value, where
$$H = 2^{110}210^{1800}\left(\prod_{p \leq 76176} p\right)^{19} + 128.$$

3. With $B = 17b$, the pole system with preparation parameter zero is
$$p^2 = B(1-e)(1+3e), \qquad Bq^2 = (s^2 - Be(B-b^2))(s^2 + 3Be(B-b^2)).$$
Its solvability excludes negative odd $v_\ell(b)$ for $\ell \notin \{2, 3, 17\}$. It is soluble whenever $0 < b < 17$, its reduced denominator is square, it is a square in $\mathbb{Q}_2, \mathbb{Q}_3, \mathbb{Q}_{17}$, and its valuations there are at least $3, 1, 2$, respectively. We retain the same restricted-converse and descent hypotheses supporting this assertion, and their simultaneous-realization construction. Lemma 2 proves the nonzero-witness consequence of these inputs.

4. The simultaneous prime-pattern input permits the source's prescribed prime values, specified Gaussian generators, and reduction-order conditions, arbitrary fixed divisibility, and arbitrary fixed odd avoidance. It applies to the finite collection of comparison scales after the root points have been fixed.

The prime-pattern input includes the stated application of Sun's theorem. No new conjecture is introduced by the height theorem used below: von Buhren's result [8] is a published theorem. Its effect here is ineffectivity of one fixed coefficient.

2 The sum of four traces controls the four labels

For nonzero integers $n_1, \ldots, n_4$ put

$$T(\mathbf{n}) = \mathrm{Tr}_{K/\mathbb{Q}}\left(\sum_{i=1}^{4} x([n_i]P)\right).$$

The aggregate trace-height theorem [2] proves that there are fixed constants $C > 0$ and $J_0 \geq 0$ such that

$$\max_i |n_i|^2 \leq Ch(T(\mathbf{n})) + J_0. \tag{1}$$

The multiplicative constant comes from fixed geometric data; the additive constant is ineffective.

For clarity, the essential hypotheses of this application are as follows. After rescaling the two conjugate curves to $E_0 : y^2 = x^3 - x$, their source points are independent over $\mathbb{Z}[i]$. The sum of their abscissas has no constant restriction to an abelian translate through a tuple of nonzero multiples. Within one block, coincident poles have tangent multipliers in the rational

ratios of their integer labels, so the leading coefficient is a nonzero sum of positive rational squares. Poles in different blocks cannot coincide by independence. The closed graph of the trace function therefore avoids the exceptional abelian translates at every source tuple. Applying von Buhren's theorem to this fixed graph and its fixed finite-rank subgroup gives (1), including a uniform additive constant. The companion paper [2] supplies the full proof over the algebraic closure, so translates defined only over an extension field are included.

3 The exact rational source

We use the point charts, local tests, and denominator guards of the established source [3]. We describe their data and the changes defining the present family.

There are nine points over K, named

$$A, S, T_1, Z, H_0, U_A, U_S, U_T, U_Z.$$

For each point retain two rational coordinates h_0, h_1 and set

$$h = h_0 + h_1\sqrt{2}, \qquad x = h^2, \qquad j^2 = h^4 - D, \qquad y = hj. \tag{2}$$

This chart describes precisely the nonzero points of $\mathbb{Z}P$. Indeed, let $R_0 = (1+\sqrt{2}, 2\sqrt{2})$, so $2R_0 = P$. The group input implies that every $Q \in E(K)$ is $mR_0 + T$ with $T \in E[2](K)$. For the standard two-isogeny descent map, $\delta(R_0) = 1+\sqrt{2}$ has opposite signs at the real embeddings, whereas the torsion classes are $1, -D, d, -d$, with equal signs. A square abscissa therefore forces m even. Of these torsion classes only 1 is square: $-D, -d$ are negative and $N_{K/\mathbb{Q}}(d) = 7$ is not a rational square. Thus $Q \in \mathbb{Z}P$. Conversely, $\delta(2R) = 1$ gives a square abscissa for every nonzero multiple of P. Neither $h = 0$ nor $j = 0$ is possible in (2). A direct proof of the descent map and all these assertions is given in [6].

The finite root j is represented by one monic quartic, without a finite choice of charts. Write

$$h^4 - D = A + B\sqrt{2}, \quad A = h_0^4 + 12h_0^2h_1^2 + 4h_1^4 - 11, \quad B = 4h_0h_1(h_0^2 + 2h_1^2) + 6.$$

Set

$$c = 2A + B, \qquad d_1 = -A - B, \qquad j' = (c + d_1\sqrt{2})j.$$

Writing $(c + d_1\sqrt{2})^2(A + B\sqrt{2}) = A' + B'\sqrt{2}$ gives

$$A' = 6A^3 - 9AB^2 - 4B^3, \qquad B' = -4A^3 + 6AB^2 + 3B^3.$$

For $j = u + v\sqrt{2}$, direct expansion gives

$$j' = 2(u^3 - 2v^3) + (4v^3 - u^3)\sqrt{2}.$$

Its rational component is nonzero, since 2 is not a rational cube and $j \neq 0$. The multiplier has norm $2A^2 - B^2$, which is nonzero for rational $(A, B) \neq (0, 0)$ because 2 is not a rational square. Thus the reconstruction uses only these two elementary rational obstructions. The general quadratic extension version is proved in [7]. We may therefore use the single chart

$$Y^4 - 4A'Y^2 + 8(B')^2 = 0, \quad Y \neq 0, \qquad j' = Y/2 + (B'/Y)\sqrt{2}, \qquad j = \frac{j'}{c + d_1\sqrt{2}}. \tag{3}$$

The point-chart factors Y, $A'^2 - 2B'^2$, $2A^2 - B^2$, and the quartic derivative are automatically nonzero at every rational retained-base specialization and every algebraic root. Indeed, $A + B\sqrt{2} = h^4 - D$ cannot vanish: taking norms would imply $N(h)^4 = 49$, which has no

rational solution. The homogeneous cubic $-4A^3+6AB^2+3B^3$ has no nontrivial rational zero, because $4T^3-6T-3$ is Eisenstein at 3. Finally,

$$A'^2-2B'^2=(2A^2-B^2)^2(A^2-2B^2)$$

and the discriminant of the quartic in (3) is

$$2^{15}B'^2(2A^2-B^2)^4(A^2-2B^2)^2\neq 0.$$

The exact square-test proof is given in [10]. These four factors per point may therefore be omitted from the finite-algebra localization product. Every other denominator of the affine point operations remains guarded.

The four root labels (n,a,t,b) are nonzero integers; no parity condition is needed for soundness. The simultaneous completeness construction retains its original odd labels. The scalar shift is now computed directly as $S-P$, in the affine difference chart. Its excluded equal-abscissa cases are $S=\pm P$, both avoided by that construction.

Put $Q_0=P$, $(Q_1,Q_2,Q_3,Q_4)=(A,S,T_1,Z)$, $U_0=H_0$, and $V_i=U_i-Q_i$. Write $z=-x/y$ and $f=1/x$. The truncated formal logarithms are

$$\ell_r(Q)=z(Q)\sum_{j=0}^{r/4-1}c_j(Df(Q)^2)^j,\quad c_0=1,\quad c_1=-2/5,\quad c_j=c_{j-1}\frac{4j-5}{4j+1}.$$

Use precisions $(12,40,40,40)$, ratio orders $(10,39,39,39)$, and scalar order 20. With $\beta=216X^4+450459$, the sixteen tests are the ten formal-group tests $f(U_i),\sigma f(V_i)$, the four differences between the paired logarithmic ratios, the direct target test for β, and the direct Lucas polynomial

$$v^2-u(u-2)vw+(u-2)^2w^2-1$$

in the three corresponding ratios. Their assigned orders are 1 for each of the first ten tests and $10,39,39,39,10,39$ for the remaining six. These tests and their construction are unchanged.

3.1 Two comparison scales

We use precisely the two positive rational square scales

$$A_0=1,\qquad A_1=\left(\mathrm{Tr}_{K/\mathbb{Q}}\sum_{i=1}^{4}x(Q_i)\right)^2. \tag{4}$$

They are nonzero: the root points are nonzero doubles and their abscissas are positive in both real embeddings. For the retained slope λ, each scale is inserted into the same normalized comparison detector of [3]:

$$t_r=A_r\lambda,\quad \delta_r=t_r-1,\quad F_r=\prod_{\theta\in\Theta}(t_r-\theta),\quad R_r=\frac{F_r}{A_r^5\delta_r^6},$$
$$\Theta=\{0,-1/2,-1,1/2399,-32/407\}.$$

$$W_r=\prod_{j=1}^{16}\left(1+R_r\left(\frac{D_j}{\delta_r^{m_j+1}R_r^{m_j}}\right)^2\right),\qquad b_r=\frac{R_r}{\mathrm{Tr}(W_r^{-1})^2}.$$

All displayed denominators and inherited guards are retained, except for the automatic point-chart factors proved above. Each b_r receives a pole certificate.

For each pole put $B = 17b$. We use the single rational chart

$$e = \frac{\tau^2 - B}{\tau^2 + 3B}, \qquad p = \frac{-4B\tau}{\tau^2 + 3B}, \qquad \tau^2 + 3B \neq 0,$$

with retained parameters τ, s, and require the remaining quadratic pole root q to be nonzero. The following lemma proves completeness of this restriction under the inherited arithmetic inputs.

Lemma 2. *Suppose that b has square reduced denominator,*

$$0 < b < 17, \qquad b \in \mathbb{Q}_2^{\times 2} \cap \mathbb{Q}_3^{\times 2} \cap \mathbb{Q}_{17}^{\times 2}, \qquad v_2(b) \geq 3, \quad v_3(b) \geq 1, \quad v_{17}(b) \geq 2.$$

The inherited pole-realization proof supplies a certificate with $pqs \neq 0$ and $e \neq 1$. Consequently the single chart above, with $q \neq 0$, suffices. The restricted predicate has the same pole obstruction, and adds no arithmetic hypothesis.

Proof. Set $V = B(B - b^2)$. Its positivity and its local square conditions are precisely those verified in the zero-preparation argument of [4]. Let β_0, θ be the positive squarefree representatives of B, V. Their supports are disjoint, $\beta_0 \equiv 17 \pmod{24}$, $\theta \equiv 1 \pmod{24}$, and both mutual local-square conditions hold.

The inherited simultaneous-realization proof in [9], with source identifier

`lem:original-seven-simultaneous-realization`,

constructs points on $E_n : y^2 = x(x - n)(x + 3n)$ and $E_{n\theta}$ with nonzero abscissas of squareclass β_0. Both are outside two-torsion. Indeed, the construction includes a fresh prime of n outside $\beta_0\theta$, so β_0 differs in squareclass from n and $n\theta$; its positive squareclass also differs from their negative multiples by -3. These are the only nonzero two-torsion abscissas. This non-torsion assertion is also explicit in the source descent proof. Thus both ordinates are nonzero.

Writing $V = \theta c^2$ and rescaling as in that proof gives

$$(B, y_1) \in E_{e_0}(\mathbb{Q}), \qquad (Bs^2, y_2) \in E_{e_0 V}(\mathbb{Q}), \qquad sy_1y_2 \neq 0.$$

The substitutions $e = e_0/B$, $p = y_1/B$, $q = y_2/(B^2 s)$ give exactly the two pole equations and $pqs \neq 0$. The first conic excludes $e = 1$. Set $\tau = p/(e - 1)$. Its equation gives

$$\tau^2 + 3B = \frac{4B}{1 - e} \neq 0,$$

and then gives the displayed reconstruction of e, p. Every restricted witness is an original pole witness, so its obstruction is unchanged. The construction establishes completeness for every input satisfying the displayed conditions. No equivalence of the two pole predicates for arbitrary rational b is asserted. □

3.2 A linear scalar normalization

The fixed scalar normalization is

$$\begin{gathered} \ell = 76249, \quad r = \beta(\beta - 1)(\beta - 2401)(32\beta + 343)(2\beta - 1), \\ R_L = \frac{102^2 \ell^L r}{(1 + C_L r)^2}, \quad C_L = 17(\ell^{(L+1)/2} + 1). \end{gathered} \tag{5}$$

Here $L \geq 3$ is odd. The scalar norm tree and its six rational component tests are unchanged: they concern $f(H_0), f(S - P), z(S - P)/z(H_0)$ with orders $1, 1, 1, 1, 20, 20$. Its output W supplies the scalar pole input $R_L W^2$.

The finite residue certificate verifies that the five contact values are not attained by $216x^4 + 450459$ modulo ℓ, and that $1 + 17r$ never vanishes there. Consequently, for every rational input,

$$v_\ell(R_L) = \begin{cases} L, & v_\ell(X) \geq 0, \\ L - 20v_\ell(X), & v_\ell(X) < 0. \end{cases} \tag{6}$$

These valuations are positive and odd. At a consumed contact prime different from ℓ, the denominator is a unit and the contact valuation is unchanged. For integral X, the reduced denominator of R_L is square, it has the required local squareclasses and depths at $2, 3, 17$, and

$$0 < R_L < \frac{36}{\ell r} < 1.$$

The scalar norm argument therefore gives

$$v_\ell(a - 1) \geq 19L + 1. \tag{7}$$

The normalization calculation in [2], together with the finite-field avoidance certificate [5], proves these statements and the preparation-zero realization. The supplied checker independently verifies every residue modulo 76249 and the normalization identity as an exact polynomial identity.

4 Soundness

The affine guard for $S - P$ excludes $a = 1$. Let $c = |a - 1| > 0$ and $B = \max(|n|, |a|, |t|, |b|)$. Equation (7) and $L \geq 3$ give $c \geq \ell^{19L+1}$ and therefore $B \geq c - 1 > 4$. In particular, the elementary residual bound $|\Delta| \leq 5B^4$ used below is valid. If the two contact radicals at the scales (4) have their kth powers dividing $N \neq 0$, the contact-height theorem gives

$$h(\lambda),\ h(A_1\lambda) \leq (H + \log 3)|N|^{19/k}.$$

Since $h(A_1) = 2h(\operatorname{Tr} \sum_i x(Q_i))$, the height inequality for a quotient and (1) give constants $K \geq 1$, $J \geq 0$, independent of X, L, such that

$$B^2 \leq K|N|^{19/k} + J. \tag{8}$$

The local formal-logarithm arguments of the source are unchanged; hence, with $\Delta = t^2 - a(a - 2)tb + (a - 2)^2b^2 - 1$, they imply

$$h(\beta) \leq Hc^{19/20}, \tag{9}$$

$$\Delta \neq 0 \Longrightarrow B^2 \leq K(5B^4)^{19/39} + J, \tag{10}$$

$$\beta \neq n \Longrightarrow B^2 \leq K(2e^{h(\beta)}B)^{19/10} + J. \tag{11}$$

Choose a fixed threshold T greater than each of

$$\begin{gathered} 16, \qquad 2\sqrt{2J} + 2, \qquad 2(2K5^{19/39})^{39/2} + 2, \\ (128H)^{20}, \qquad 8\big(10\log(2K) + 19\log 2\big). \end{gathered} \tag{12}$$

For $c > T$, one has $B \geq c - 1 > c/2$ and $B^2 > 2J$. If $\Delta \neq 0$, equation (10) gives $B^{2/39} \leq 2K5^{19/39}$, contrary to (12). Thus $\Delta = 0$. The elementary Lucas growth lemma proved in [3] then gives $\log B > c - 4$. If $\beta \neq n$, equation (11) gives

$$\log B \leq 10\log(2K) + 19\log 2 + 19h(\beta) < \frac{c}{8} + \frac{19c}{128} < c - 4,$$

using (9) and the threshold. This contradiction proves $\beta = n \in \mathbb{Z}$. Finally the elementary input-map lemma of [3] says that $216X^4 + 450459 \in \mathbb{Z}$ for rational X if and only if $X \in \mathbb{Z}$.

By (7), every source solution has $c \geq \ell^{19L+1}$. Therefore every odd L with $\ell^{19L+1} > T$ satisfies the soundness conclusion. The constant T is fixed but ineffective because J is ineffective.

5 Completeness

Fix an integer input X and a fixed odd L. All numerator and denominator primes of (5) form a finite set. The source realization first chooses the odd root labels with the scalar label $a - 1$ sufficiently divisible and deep at these primes, and with the corresponding points in the required real neighborhoods. The explicit Lucas construction then gives odd nonzero t, b with $\Delta = 0$ and arbitrarily large growth. The input label is chosen to be $n = \beta$.

At this stage the two scales in (4) are fixed positive rational squares. Apply the same simultaneous prime-pattern theorem to these two scales. It chooses the slope and contact primes with the prescribed conjugate reduction orders while avoiding the previously fixed odd modulus. The subsequent Chinese remainder choice of the auxiliary index h and the points $U_i = [hn_i]P$ supplies the sixteen local tests to their assigned precisions and preserves the fixed scalar requirements and real neighborhoods. This is precisely the simultaneous construction of [4, 3]; its prime-pattern input accepts any fixed finite collection of the positive square scales and any fixed finite avoidance modulus.

Each of the two comparison inputs is thereby prepared for the zero-preparation pole theorem. The scalar input is prepared as well: R_L has square reduced denominator and the required local squareclasses, the six normalized scalar tests can be made arbitrarily small at the fixed finite places, and the scalar norm output can be made arbitrarily close to 1 in the real absolute value. Thus $0 < R_L W^2 < 17$, with the requisite denominator and local conditions. The three pole certificates exist by the inherited completeness input. The direct point chart (3) covers all nine chosen points; its multiplier needs no further choice. Lemma 2 applies independently to each pole input. Its parameters belong only to that pole, so the certificates can be chosen simultaneously without any common-parameter assumption. All pole roots are nonzero and the single conic chart applies. All remaining inherited guards are maintained by the same realization.

This proves completeness for every fixed odd $L \geq 3$. Together with the preceding section it proves the arithmetic assertion for every sufficiently large such L.

6 Elimination of finite roots and the uniform degree

The retained coordinates and finite roots are as follows.

Retained rational coordinates	Count
Input	1
Nine square-root-abscissa pairs	18
Comparison slope	1
Two comparison-pole parameter pairs	4
Scalar-pole parameter pair	2
Total before finite-algebra elimination	26

There are nine quartic point roots and three quadratic pole roots. The finite-algebra rank is $4^9 2^3 = 2^{21}$. Each pole root is already guarded nonzero in the source, so every zero stratum is inconsistent and is discarded. The first conics use one chart. The theorem of [1] eliminates all

twelve finite roots and retains every nonautomatic guard using one rational coordinate. This gives precisely 27 total polynomial variables.

Write B_0 for the algebra of the nine point roots and A_0 for the algebra after the quadratic pole roots have been adjoined. These algebra symbols are unrelated to the earlier point-chart coefficients. Put $b_0 = 4^9 = 2^{18}$ and $n_0 = \dim A_0$. If the pole roots are $q_1, \dots, q_s$, their relations are $q_j^2 = c_j$ with $c_j \in B_0$. Exact coefficient-dependency inspection shows that the source guard is $G = gq_1 \cdots q_s$ with $g \in B_0$. There are no closed residual equations. Define $T = g^2 \prod_j c_j$ and

$$\det_{B_0}(I - vM_T) = \sum_{j=0}^{b_0} a_j v^j, \qquad a_0 = 1.$$

Let $\theta_1, \dots, \theta_m$ be the point and pole generators in order, and let δ_i be the discriminant of the ith point quartic. They are scalar polynomials, nonzero at every rational base point by the point-chart proof above. Put

$$R = n_0 \left(1 + \sum_{i=1}^{m} \sum_{u,v=1}^{n_0} (M_{\theta_i}^{A_0})_{uv}^2 + \sum_{u,v=1}^{b_0} (M_g^{B_0})_{uv}^2 \right).$$

For each $1 \le r \le b_0$, set

$$c = a_r, \qquad h_r(Z) = \sum_{j=0}^{r} a_j Z^{r-j}, \qquad E_r = c - h_r(T),$$

$$D_r = c^2 \prod_{i=1}^{9} \delta_i^2, \qquad S_r = r(s+1) + 108,$$

$$K_r = (2R)^{S_r}(1 + c^2) \prod_{i=1}^{9} (1 + \delta_i^2), \qquad C_r = D_r + 4RK_r,$$

$$\beta_r = D_r E_r \left(C_r^m + \sum_{i=1}^{m} D_r^i C_r^{m-i} \theta_i \right), \qquad H_r(U) = \det_{A_0}(I - UM_{\beta_r}).$$

The required polynomial is the following finite integer product:

$$F = H_{b_0}(U) \prod_{r=1}^{b_0 - 1} \left(H_r(U)^2 + \sum_{j=r+1}^{b_0} a_j^2 \right).$$

Every operation has an explicit finite range. There is exactly one extra rational coordinate, namely U.

The separating-element theorem of [1] proves that this polynomial has precisely the guarded source projection. Here is the reason for its treatment of every specialization. The actual degree of $\det_{B_0}(I - vM_T)$ counts the point-algebra characters with $T \neq 0$. At that degree, E_r/c is the idempotent of those factors. Moreover

$$|c| = \prod_{\chi : T(\chi) \neq 0} |g(\chi)|^2 \prod_j |c_j(\chi)|.$$

Since all generator and g-values have absolute value at most R, isolating a factor bounds each required quadratic root separation from below. The scalar discriminants similarly bound the point-root separations. Thus D_r/K_r is a valid positive separation bound, and the powers of C_r/D_r distinguish the characters where the guard is nonzero. Multiplication by E_r annihilates the other factors, including their nilpotents. A rational root of H_r therefore gives a one-dimensional rational simultaneous eigenspace of all multiplication matrices, which reconstructs

a guarded rational character. Conversely any such character gives a rational root. If $c = 0$, then $H_r = 1$; the squared tail coefficients exclude every other incorrect degree case. This also covers complex residue factors and an empty guarded part. The full proof and coefficient estimates are in [1].

For clarity, if the certified weight and logarithmic operator norm of β_r are at most $W_1 r + W_0$ and $L_1 r + L_0$, respectively, the principal-minor estimates give

$$\begin{aligned}\deg F &\le n_0(W_1 b_0^2 + (W_0+1)(2b_0-1)),\\ \log_2 \|F\|_1 &\le n_0(L_1 b_0^2 + (L_0 + \lceil \log_2 n_0 \rceil + 1)(2b_0-1))\\ &\quad + (b_0-1)\lceil \log_2(b_0+1)\rceil.\end{aligned}$$

The proof sums every degree case and uses $2\sum_{r=1}^{b_0-1} r + b_0 = b_0^2$.

Here $m = 12$, $s = 3$, and $n_0 = 2^{21}$. The point-generator weights are all 6; the remaining weights are $1568138, 1935231, 116318$. The complete guard and its point factor have weights 4423559 and 803872. The resulting linear majorants are

$$W_1 = 772159866, \qquad W_0 = 14534041359,$$

so

$$\deg F \le 111295944630211471361441792 < 2^{87}.$$

All degree weights are independent of the fixed exponent L. The source uniform-degree checker replaces ℓ^L and $\ell^{(L+1)/2}$ by independent coefficient symbols, each of weight zero. Their integer specialization cannot increase a source weight. The same bound therefore holds for every fixed odd L. No coefficient-height bound for a member certified to exceed the ineffective threshold is inferred.

7 Reproduction and the status of a numerical example

The directory `trace27_nontorsion` contains the exact guarded source, its builder, independent source checks, and its uniform-degree certificate. The source command

```
python3 reproduce.py --anchor-exponent 137731
```

reconstructs and checks the numerical source. The exponent is a fixed coefficient choice and does not add a quantified variable. Its source has 7369 coefficient nodes; integer coefficient conversion gives 3032 nodes. Source verification covers nine quartic point charts, three rational-square clearings, all 104 atomic source guard factors, and the 36 automatic factors omitted from the guard. The scalar normalizer is checked at all 76249 residue classes.

The directory `active_elimination` contains the compiler and independent checker:

```
build_active_specification.py
check_active_specification.py
check_active_degenerations.py
```

The final specification is in `active27`. The compiler uses the verified source coefficient graph and writes exactly the determinant and finite product defined above. The independent checker verifies the homogeneous guard and every coefficient dependency, certifies companion-column norm inequalities, and sums every degree case literally. The separate degeneration checker uses determinants, Newton traces, and exact polynomial gcds to cover inactive nilpotents, complex residue factors, and disappearing branches. All programs require only the Python standard library.

For the numerical exponent 137731, the resulting coefficient height is bounded by 2^{C_*} with

$$C_* = 217401420128398649581869203437 < 2^{98}.$$

This is a bound for a specified algebraic family member. No argument here certifies that this numerical exponent exceeds the ineffective sufficient threshold. The defining-polynomial assertion is the existence theorem for every sufficiently large odd exponent, with the uniform degree bound proved above.

References

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[9] E. Glazer, *Diophantine definition of the integers over the rationals*, Version 9, `manuscripts/quartic_v9.tex`, commit `14e8f601fe0d9173d4a5f084aedc3e07066a7688`. https://github.com/elliotglazer/diophantine-z-q. The inherited restricted-converse correction is identified in [4].

[10] *Optimal polynomial square tests in quadratic number fields*, companion paper, 8 November 2026, `quadratic_square_predicates_20261108_v10.tex`.